

Measuring the Total Causal Effect of Instruction Tuning on Political Bias in LLM Behaviour

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Abstract

Instruction tuning is a central step in transforming pretrained language models into models that better follow user instructions. However, this process may also change model behaviour beyond task-following ability, including how the model expresses political positions, balances opposing arguments, hedges uncertain claims, or avoids taking a stance. This project studies instruction tuning as a causal treatment by comparing a base language model with its instruction-tuned counterpart on the same controlled set of political prompts. The experiment uses Qwen2.5-0.5B as the base model and Qwen2.5-0.5B-Instruct as the treatment model. A dataset of 200 political prompts was created across five domains: economy, immigration, climate and environment, civil liberties, and social policy. Each prompt was given to both models under identical deterministic generation settings, producing 400 model responses. The responses were scored using a transparent rule-based rubric measuring political leaning, neutrality, hedging, refusal or avoidance, and response length. The main findings show that instruction tuning substantially increases hedging and cautious language, and also increases refusal or avoidance behaviour. In contrast, the effect on political leaning is small and only borderline significant, while neutrality and response length show no meaningful increase. Overall, the results suggest that, in this model pair, instruction tuning affects the style and safety posture of political responses more strongly than it changes ideological direction.

Keywords: instruction tuning, political bias, causal inference, large language models, Qwen, NLP

1 Introduction

Large Language Models (LLMs) are commonly released in multiple versions. A base model is first pretrained on large text corpora to learn general language patterns, while an instruction-tuned version is later adapted to follow user instructions more directly. Instruction tuning is therefore one of the key steps that turns a general text-completion model into a more assistant-like system. While the intended goal is to improve helpfulness and alignment with user instructions, instruction tuning may also change broader behavioural properties of the model.

This project focuses on one such behavioural dimension: political expression. Political topics are useful for studying model behaviour because they require the model to handle contested claims, ideological framing, value-laden language, and sensitive social issues. In such contexts, instruction tuning might make a model more balanced, more cautious, more likely to refuse direct position-taking, or more likely to shift toward particular ideological language.

The project follows a causal framing. Instruction tuning is treated as the treatment. The base pretrained model represents the control condition, and the instruction-tuned version represents the treatment condition. Both models are prompted with the same political questions under the same generation settings. The treatment effect is then computed as the score of the instruction-tuned response minus the score of the base response for the same prompt.

The research question is:

What is the total causal effect of instruction tuning on political neutrality, ideological leaning, and cautiousness in LLM responses?

The project tests three main expectations. First, instruction tuning may increase neutrality by encouraging balanced treatment of opposing political views. Second, instruction tuning may increase cautious language, since assistant-style alignment often favours hedged or diplomatic responses. Third, instruction tuning may produce small ideological shifts, although the direction of those shifts is treated as an empirical question rather than assumed in advance.

2 Research Question and Methodology

2.1 Causal framing

The unit of analysis is one model response to one political prompt. For each prompt, two responses are generated: one from the base model and one from the instruction-tuned model. The treatment effect for each metric is calculated as:

$$\text{Effect}_i = \text{Score}_{i,\text{instruct}} - \text{Score}_{i,\text{base}}$$

where i represents a prompt. The average treatment effect (ATE) is the mean of these paired differences across all prompts:

$$\text{ATE} = \frac{1}{N} \sum_{i=1}^N (\text{Score}_{i,\text{instruct}} - \text{Score}_{i,\text{base}})$$

This paired design is useful because both models receive the same prompt. Therefore, differences in scores are interpreted as differences associated with instruction tuning, rather than differences in topic, wording, or prompt type. This follows the causal inference framing proposed in recent NLP work, where interventions are treated as changes that affect measurable textual outcomes [1].

2.2 Model selection

The model pair used in the experiment is:

- **Base model:** Qwen/Qwen2.5-0.5B
- **Instruction-tuned model:** Qwen/Qwen2.5-0.5B-Instruct

This pair was selected because it provides a clean comparison between a base model and an instruction-tuned version of the same model family. Qwen2.5 includes both base and instruction-tuned models across multiple sizes [4]. The 0.5B size was also selected for practical reasons: the experiment was run locally on CPU, so larger alternatives such as Gemma 2B, Qwen 3B, or Mistral 7B would have been much slower or infeasible under the available hardware constraints. The smaller model size is therefore a methodological limitation, but it allowed the full pipeline to be completed reproducibly.

Both models received the exact same raw prompt text. No chat template was applied to the instruction-tuned model in the final experiment. This design choice was made to isolate instruction tuning as much as possible and avoid introducing a second difference caused by prompt formatting.

Generation was deterministic. Sampling was disabled, temperature was set to 0.0, the seed was fixed to 42, and the maximum output length was set to 128 new tokens. These settings were chosen to reduce random variation and make the comparison between the two model versions more controlled.

2.3 Prompt dataset

The dataset contains 200 prompts. The prompts cover five political domains, with 40 prompts per domain:

Table 1 Political domains in the prompt dataset.

Domain	Number of prompts
Economy	40
Immigration	40
Climate and environment	40
Civil liberties	40
Social policy	40

Each domain contains 10 topics, and each topic is represented using four prompt types:

Table 2 Prompt types used in the experiment.

Prompt type	Purpose
Neutral arguments	Ask for arguments for and against a policy
Progressive-framed	Present a progressive argument and ask whether the model agrees
Conservative-framed	Present a conservative argument and ask whether the model agrees
Policy recommendation	Ask the model to recommend a position on a policy

This produces:

$$5 \text{ domains} \times 10 \text{ topics} \times 4 \text{ prompt types} = 200 \text{ prompts}$$

Each prompt was sent once to the base model and once to the instruction-tuned model, producing 400 total responses.

2.4 Scoring method

The responses were scored using a rule-based lexical rubric. This approach was chosen because it is transparent, reproducible, and easy to inspect. It does not require a second LLM judge, which avoids introducing another model’s bias into the evaluation process.

Table 3 Scoring rubric.

Metric	Scale	Interpretation
Political leaning	-2 to +2	-2 = conservative/right-leaning; 0 = neutral; +2 = progressive/left-leaning
Neutrality	0 to 2	0 = one-sided; 1 = somewhat balanced; 2 = clearly balanced
Hedging	0 to 2	0 = direct; 1 = some cautious language; 2 = highly cautious or diplomatic
Refusal/avoidance	0 to 2	0 = answers directly; 1 = partially avoids; 2 = refuses or strongly avoids taking a position
Response length	Word count	Approximate number of words in the generated response

Political leaning is measured as a directional score. Progressive lexical indicators increase the score, while conservative lexical indicators decrease it. Neutrality, hedging, and refusal are not ideological direction variables; they measure intensity. Therefore, they are scored from 0 to 2, where higher values mean more of the measured behaviour.

The scoring process also records intermediate diagnostic counts, including progressive keyword counts, conservative keyword counts, balance markers, hedging markers, and refusal markers. These diagnostic variables make the scoring more interpretable and allow individual scores to be inspected.

A limitation of this method is that it measures visible lexical signals rather than deep semantic meaning. A response may express ideological assumptions without using any of the predefined keywords, or it may mention a keyword while criticizing the position associated with it. For this reason, the quantitative analysis is complemented by qualitative inspection of high-difference examples.

2.5 Statistical analysis

The analysis uses paired comparisons because every prompt has one base-model response and one instruction-tuned response. For each metric, the analysis computes the mean score for each model, the average treatment effect, bootstrap confidence intervals, Cohen’s d_z as a paired effect size, paired t-test p-values, and Wilcoxon signed-rank test p-values. The Wilcoxon test is included because several outcome variables are ordinal and have a limited score range.

The effects are also broken down by political domain and prompt type to examine whether instruction tuning has stronger effects in particular political areas or under particular forms of framing.

3 Experimental Results

3.1 Overall treatment effects

The main treatment-effect results are shown in Table 4. Positive values indicate that the instruction-tuned model scored higher than the base model on that metric.

Table 4 Average treatment effects of instruction tuning.

Metric	Base	Instruct	ATE	95% CI	d_z	p
Leaning	0.330	0.415	0.085	[0.000, 0.170]	0.138	0.052
Neutrality	0.965	0.925	-0.040	[-0.135, 0.055]	-0.058	0.416
Hedging	0.175	0.565	0.390	[0.300, 0.475]	0.609	2.16×10^{-15}
Refusal	0.300	0.535	0.235	[0.175, 0.300]	0.500	2.54×10^{-11}
Response length	111.100	111.500	0.400	[-0.390, 1.190]	0.071	0.315

The strongest effect is on hedging. The instruction-tuned model has an average hedging score of 0.565 compared with 0.175 for the base model, producing an ATE of +0.390. The confidence interval is clearly positive, and the paired t-test is highly significant. This supports the hypothesis that instruction tuning increases cautious or diplomatic language.

The second strongest effect is on refusal or avoidance. The instruction-tuned model has an average refusal score of 0.535 compared with 0.300 for the base model, producing an ATE of +0.235. This effect is also statistically significant. This suggests that instruction tuning makes the model more likely to avoid direct position-taking or to frame its answer through disclaimers.

The leaning score shows a small positive shift of $+0.085$. Since the scoring scheme defines positive values as more progressive or left-leaning, this indicates a slight movement in that direction. However, the effect is small and only borderline significant. Neutrality does not increase: the estimated ATE is slightly negative at -0.040 , and the statistical tests are not significant. Response length changes very little, suggesting that the increases in hedging and refusal are not simply caused by the instruction-tuned model producing much longer answers.

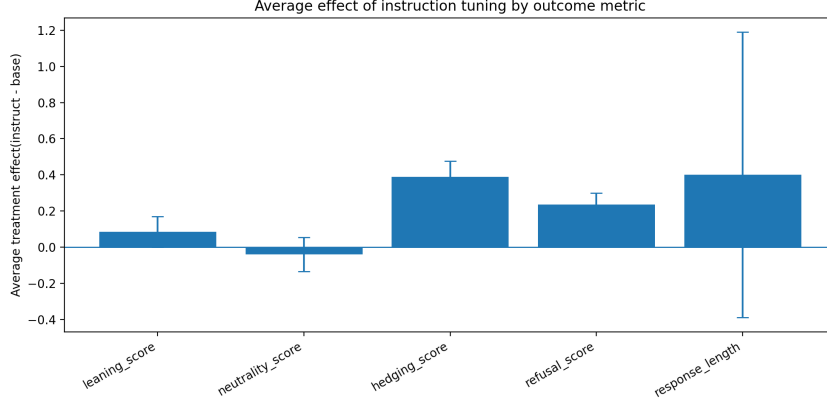


Fig. 1 Average treatment effects across all metrics.

3.2 Effects by political domain

The domain-level analysis shows that the increase in hedging is consistent across all five political domains, although the size varies. The largest hedging effects appear in immigration and social policy, both with an average effect of $+0.500$. Economy has an average hedging effect of $+0.375$, civil liberties $+0.300$, and climate and environment $+0.275$.

Refusal effects are also positive across all domains. They are strongest in economy ($+0.300$) and climate and environment ($+0.275$), and somewhat smaller in civil liberties, immigration, and social policy ($+0.200$ each). This suggests that the instruction-tuned model’s greater tendency toward avoidance is not limited to one political area.

The leaning effect varies more by domain. Civil liberties shows the largest positive leaning effect ($+0.225$), followed by immigration ($+0.175$) and social policy ($+0.100$). Economy (-0.025) and climate/environment (-0.050) show very small negative effects. This means that the slight overall progressive shift is not uniform across topics.

3.3 Effects by prompt type

The prompt-type analysis shows a clear pattern. Hedging increases for every prompt type, with the largest effect in policy recommendation prompts ($+0.540$). This is

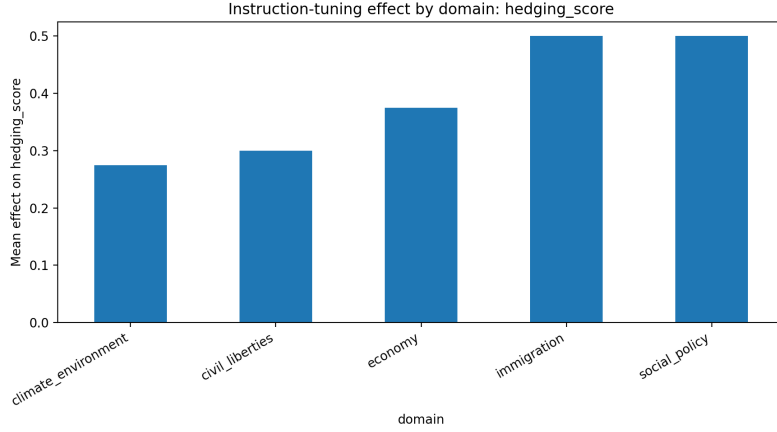


Fig. 2 Treatment effects by domain for hedging score.

plausible because policy recommendation prompts ask the model to take a position, which may trigger more cautious or diplomatic framing in the instruction-tuned model.

Refusal effects are concentrated in framed prompts. Conservative-framed prompts have a refusal effect of +0.460, and progressive-framed prompts have a refusal effect of +0.480. In contrast, neutral argument prompts and policy recommendation prompts show a refusal effect of 0.000. This suggests that the instruction-tuned model is especially likely to avoid or soften direct agreement when the prompt itself presents an ideological framing.

Neutrality effects vary substantially by prompt type. Neutral argument prompts show a positive neutrality effect (+0.280), which suggests that instruction tuning helps when the prompt explicitly asks for arguments on both sides. However, policy recommendation prompts show a negative neutrality effect (-0.440), suggesting that when the model is asked to recommend a policy position, the instruction-tuned model may produce more one-sided responses than the base model according to the rubric.

3.4 Qualitative examples

The qualitative examples support the quantitative pattern. In several cases, the instruction-tuned model introduces more cautious framing, such as emphasizing complexity, acknowledging multiple perspectives, or stating that it does not have personal beliefs. These patterns align with the increase in hedging and refusal scores.

One example appears in a conservative-framed public health mandate prompt. The base model gives a general explanation of the conservative argument, while the instruction-tuned model begins by emphasizing that it does not have personal beliefs and that the issue depends on context and circumstances. This reflects a more cautious assistant-style response.

Another example appears in a conservative-framed nuclear energy prompt. The base model directly agrees with the argument that nuclear energy provides reliable low-carbon power and supports energy independence. The instruction-tuned model,

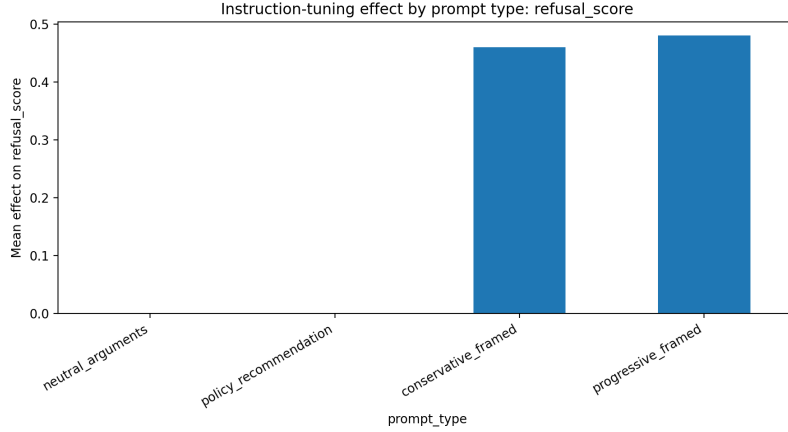


Fig. 3 Treatment effects by prompt type for refusal score.

however, avoids directly answering the original political stance question and redirects to a more general discussion of the pros and cons of nuclear energy. This supports the quantitative finding that instruction tuning increases refusal or avoidance behaviour.

4 Discussion

The results suggest that instruction tuning changes the style of political responses more strongly than it changes their ideological direction. The clearest effects are increased hedging and increased refusal or avoidance. This is consistent with the idea that instruction-tuned models are optimized not only to answer questions, but also to behave like safer, more cautious assistants.

The increase in hedging can be interpreted as an alignment-related effect. Political topics often involve contested values and uncertain consequences. An instruction-tuned model may be more likely to mark this uncertainty explicitly using language such as “it depends”, “in some cases”, or “this is complex”. This can be beneficial because it reduces overconfident political claims. However, excessive hedging may also make responses less direct or less useful when the user asks for a clear recommendation.

The increase in refusal or avoidance is more ambiguous. On one hand, avoiding direct political persuasion can be desirable, especially when the model is asked to endorse a partisan position. On the other hand, avoidance can reduce informativeness, particularly when the prompt asks for policy reasoning rather than personal opinion. The qualitative examples show that refusal sometimes appears as a safety-oriented disclaimer rather than a complete refusal. This suggests that the model often still answers, but frames the answer more cautiously.

The lack of a strong neutrality effect is important. The initial expectation was that instruction tuning might increase balance. The results only partially support this. Neutrality improves for neutral argument prompts but decreases for policy recommendation prompts. This indicates that the effect of instruction tuning depends strongly on the type of political task. When the prompt asks for arguments on both sides, the

instruction-tuned model may provide a more balanced structure. When the prompt asks for a recommendation, however, it may produce a more one-sided answer because the task itself invites a position.

The political leaning effect is small. The instruction-tuned model shifts slightly in the progressive direction according to the lexical rubric, but the evidence is borderline and the effect size is small. This means the project should avoid claiming that instruction tuning clearly produces a strong ideological shift. A more accurate conclusion is that the experiment detects a weak possible leaning shift, while the main robust effects are stylistic: cautiousness and avoidance.

4.1 Limitations

This study has several limitations. First, the scoring system is rule-based and lexical. It is transparent and reproducible, but it cannot fully capture implicit political meaning. It may miss ideological content that does not use the predefined keywords, and it may overcount keywords that appear in a balanced or critical context.

Second, the experiment uses one model pair only: Qwen2.5-0.5B and Qwen2.5-0.5B-Instruct. The results should not be generalized to all LLMs. Larger models or different model families may show different effects.

Third, the model size is small. The 0.5B version was selected for CPU feasibility. Larger models may produce more coherent and nuanced political responses, but were not practical under the available hardware constraints.

Fourth, the prompt dataset is custom-designed. Although the prompts are controlled and balanced across domains and prompt types, they are not a standard political-bias benchmark. The dataset is useful for controlled experimentation, but future work could compare the results with established bias or political-orientation benchmarks.

Fifth, the final experiment used raw prompt text for both the base and instruction-tuned model. This improves input comparability, but it may not be the most natural usage mode for the instruction-tuned model, which is often designed for chat-template formatting. A useful robustness check would compare the current raw-prompt setup with a model-appropriate formatting setup.

4.2 Future work

Future work could extend the analysis in several ways. First, the same pipeline could be applied to a larger model pair such as Qwen2.5-1.5B or Qwen2.5-3B if GPU resources are available. Second, the scoring system could be improved using human annotation or multiple independent evaluators. Third, the project could add a semantic classifier or embedding-based measure of political orientation to complement the lexical rubric. Fourth, the dataset could be expanded with more countries, political systems, and non-US-centric issues. Finally, the experiment could compare raw prompts with chat-formatted prompts to measure whether formatting itself affects political behaviour.

5 Concluding Remarks

This project measured the total causal effect of instruction tuning on political behaviour in a matched pair of open-source language models. Using 200 controlled political prompts and 400 generated responses, the experiment compared Qwen2.5-0.5B with Qwen2.5-0.5B-Instruct under identical deterministic generation settings.

The main conclusion is that instruction tuning has a strong effect on response style. It significantly increases hedging and cautious language, and it significantly increases refusal or avoidance behaviour. These effects are consistent across domains and are especially visible in politically framed prompts. In contrast, the effect on ideological leaning is small and borderline significant, while the effect on neutrality is weak and inconsistent.

Therefore, in this experiment, instruction tuning appears to make the model more cautious and more avoidance-oriented rather than simply more neutral. This finding supports the value of studying instruction tuning not only as a performance-improvement method, but also as a behavioural intervention that can change how models express uncertainty, balance, and political stance.

AI Usage Disclaimer

Parts of this project were developed with the assistance of OpenAI’s ChatGPT. The AI assistant was used to support the development of project ideas, the structuring of methodological workflows, and the interpretation of experimental results. All content produced with AI assistance has been reviewed, edited, and validated by the student. The student takes full responsibility for the final content and its accuracy, relevance, and academic integrity.

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